

Europass Curriculum Vitae



Personal information

Name / Surname
Professional Email
Home page
Nationality

Matteo Teodori

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<https://matteoteodori.github.io/>

Italian

Research Interests

Analytical and numerical models to study the structural, spatial, kinematic and thermal properties and evolution of collisional and partially collisional physical systems.

I am interested in planetary science modeling, mainly the emission of volatiles from planetary surface and subsurface with analytical and numerical Lagrangian approaches (Smoothed Particle Hydrodynamics), icy moon plumes (mainly Europa and Enceladus), and the deposition and sublimation of volatiles on the surface.

I am also interested in stellar system dynamical evolution and stability, mainly Globular Clusters and their Multiple Populations, characterized by means of analytical models and N-body simulations.

I am generally curious about fundamental physics and astrophysics, planetary sciences, stellar formation and evolution, dark matter and cosmology, numerical tools and data analysis techniques.

Research experiences

1st Dec. 2025 - Now

Post-doc fellowship at INAF - IAPS for the “Study of volatile emissions from planetary surfaces with the use of numerical methods, in the context of lunar, Martian and icy moon exploration of the Solar System” concerning the projects “Martian exploration” and “JUICE mission”.

1st June 2022 - 30th Nov. 2022

Studentship at INAF - IAPS entitled “Activity of study and formation of planetary structures, through modeling and/or remote sensing and/or laboratory data” concerning the projects “ExoMars”, “Dawn” and “TRIS”. Study of the Smoothed Particle Hydrodynamics numerical method for the simulation of volatiles emission from planetary surfaces of interest for the mentioned projects.

1st Feb. 2022 - 31th May 2022

Internship at INAF - Astronomical Observatory of Rome, finalized at learning the mathematical techniques of the “Information Field Theory” and their application to astronomical data, with reference to the high-contrast images produced within the SHARK-VIS project, an instrument intended for the LBT telescope for deep detection of exoplanets through direct images.

1st Dec. 2021 - 30th Nov. 2022

Research collaborator at the Physics Department - University of Rome “La Sapienza”, concerning the study of multi-mass dynamical models for Globular Clusters.

Education

13th Jan. 2026

PhD in Mathematics, Physics and application for Engineering, University of Campania “Luigi Vanvitelli” (XXXVIII cycle) in collaboration with **INAF - Osservatorio Astronomico d’Abruzzo**. Thesis title: [Stellar and dynamical evolution in Globular Clusters](#). Supervisors: Prof. Oscar Straniero & Prof. Lucio Gialanella.

16th Nov. 2021

Master degree in Astronomy & Astrophysics, University of Rome “La Sapienza”, *cum laude*. Thesis title: Gravothermal catastrophe in models of Globular Clusters with a mass distribution. Supervisor: Prof. Marco Merafina.

1st Oct. 2019

Bachelor’s degree in Physics, University of Rome “La Sapienza”. Dissertation title: Carbon ignition curves for massive stars. Supervisor: Prof. Oscar Straniero

Publications

2026

M. Teodori, L. Maggioni, G. Magni, M. Formisano, M. C. De Sanctis, F. Altieri, “The exogenic contribution to lunar water ice: modeling a cometary post-impact plume through Smoothed Particle Hydrodynamics” , in preparation

2026

M. Teodori, O. Straniero, M. Merafina, L. Gialanella, “[Primordial segregation in globular clusters: The degree of mass segregation as determined through dynamical models](#)”, Astronomy & Astrophysics, 707, A311, DOI: 10.1051/0004-6361/202557469.

2026

M. Teodori, L. Maggioni, G. Magni, M. Formisano, M. C. De Sanctis, F. Altieri, “[Plumes emission from a fracture on a planetary surface using Smoothed Particle Hydrodynamics: the case of Enceladus](#)”, Icarus, 443, 116765, DOI: 10.1016/j.icarus.2025.116765.

2024

M. Teodori, O. Straniero, M. Merafina “[Energy equipartition in Globular Clusters through the eyes of dynamical models](#)”, Astronomy & Astrophysics, 691, A202, DOI: 10.1051/0004-6361/202450783, [supplementary material](#).

2022

M. Merafina and **M. Teodori**, “[Generalization of the Fokker-Planck equation for stellar orbit diffusion in multi-mass star systems](#)”, arXiv: 2205.10209, DOI: 10.48550/ARXIV.2205.10209.

As a co-author

2026

L. Maggioni, **M. Teodori**, G. Magni, M. Formisano, M. C. De Sanctis, F. Altieri, “[Smoothed particle hydrodynamics modeling of volatile emissions: Drill-induced releases in the martian subsurface](#)”, Planetary and Space Science, 272, 106244, DOI: 10.1016/j.pss.2026.106244.

2025

M. Formisano, A. Raponi, **M. Teodori**, S. Bertoli, M. Ciarniello, S. De Angelis, M.C De Sanctis, G. Filacchione, A. Frigeri, P.O. Hayne, L. Maggioni, G. Magni, K.C. Sorli, “[The Role of Self-Heating and Roughness in Micro Cold Trap Stability: Implications for Lunar Poles](#)”, Monthly Notices of the Royal Astronomical Society, 543, 4187, DOI: 10.1093/mnras/staf1670.

2025

L. Maggioni, **M. Teodori**, G. Magni, M. Formisano, M.C. De Sanctis, F. Altieri, “[Gas mixing through a Smoothed Particle Hydrodynamics approach](#)”, Monthly Notices of the Royal Astronomical Society, 543, 3310, DOI: 10.1093/mnras/staf1300.

2024

M. Formisano, M. C. De Sanctis, S. Boazman, A. Frigeri, D. Heather, G. Magni, **M. Teodori**, S. De Angelis, M. Ferrari, “[Thermal modelling of the lunar South Pole: Application to the PROSPECT landing site](#)”, Planetary and Space Science, 251, 105969, DOI: 10.1016/j.pss.2024.105969.

Proceedings

2025

M Merafina, **M. Teodori**, O. Straniero, “Energy Equipartition in Globular Clusters”, Proceedings of Science, Multifrequency Behaviour of High Energy Cosmic Sources - XV (MULTIF2025), 9-14 June 2025, Palermo, Italy.

Abstract as first author

Invited talks

15-20 June 2026

M. Teodori, M. Merafina, O. Straniero, “[Dynamical evolution of Globular Clusters: the degree of mass segregation from birth to nowadays](#)”, Frontier Research in Astrophysics Workshop [FRAPWS2026–V edition](#), Palermo, Italy.

Talks

17-19 Feb. 2026

M. Teodori, L. Maggioni, G. Magni, M. Formisano, M. C. De Sanctis, G. Piccioni, “[Investigating Europa plumes with Smoothed Particle Hydrodynamics: preliminary analysis and simulation setup](#)”, MAJIS STM#10, Berlin, Germany.

1-5 July 2025	M. Teodori , L. Maggioni, G. Magni, M. Formisano, M. C. De Sanctis, F. Altieri, E. D'Aversa, M. Ciarniello, " <i>A Smoothed Particle Hydrodynamics model for volatiles emission: simulations of Enceladus' plumes</i> ", Biennial European Astrobiology Conference BEACON 2025, Reykjavik, Iceland.
3-7 Feb. 2025	M. Teodori , L. Maggioni, G. Magni, M. Formisano, M. C. De Sanctis, F. Altieri, E. D'Aversa, " <i>Modeling volatiles emission through Smoothed Particle Hydrodynamics: Enceladus' plumes</i> ", XX Congresso Nazionale di Scienze Planetarie 2025 (CNSP2025), Pescara, Italy.
8-13 Sept. 2024	M. Teodori , L. Maggioni, G. Magni, M. Formisano, M. C. De Sanctis, F. Altieri, E. D'Aversa, " <i>Volatiles emissions from surface fractures: Enceladus' plumes through Smoothed Particle Hydrodynamics simulations</i> ", Europlanet Science Congress 2024, Berlin, Germany, EPSC2024-55 , DOI: 10.5194/epsc2024-55.
19-21 Mar. 2024	M. Teodori , G. Magni, M. Formisano and L. Maggioni, " <i>Advancements in SPH modeling for volatiles emission</i> ", ISSI International Team Meeting, Bern, Switzerland.
16-20 Oct. 2023	M. Teodori , O. Straniero, M. Merafina and L. Gialanella, " <i>Dynamical evolution of Multiple Populations in Globular Clusters</i> ", STARS Across the Universe, INAF - Osservatorio Astronomico di Capodimonte, Napoli, Italy, DOI: 10.5281/ZENODO.10105297.
6-10 Feb. 2023	M. Teodori , G. Magni, M. Formisano, M. C. De Sanctis and F. Altieri, " <i>Volatiles emission from a fracture on a planetary surface: a Smoothed Particle-Hydrodynamics approach</i> ", XVIII Congresso Nazionale di Scienze Planetarie, Perugia, Italy.
14th Nov. 2022	M. Teodori , " <i>Multi-mass collisional stellar systems models for Globular Clusters</i> ", G11 Workshop, Physics Department, University of Rome "La Sapienza".
<i>Posters</i>	
29 June - 3 July 2026	M. Teodori , O. Straniero, M. Merafina, " <i>The dynamical evolution of globular clusters and their multiple populations: characterization of energy equipartition and mass segregation</i> ", European Astronomical Society Annual Meeting, Lausanne, Switzerland. Virtual participation.
2-6 Feb. 2026	M. Teodori , L. Maggioni, G. Magni, M. Formisano, M. C. De Sanctis, F. Altieri, E. D'Aversa, M. Ciarniello, G. Piccioni, " <i>Plumes with Smoothed Particle Hydrodynamics: From Enceladus to Europa and beyond</i> ", CNSP26, Reggio Emilia, Italy.
25-30 Jan. 2026	M. Teodori , L. Maggioni, G. Magni, M. Formisano, M. C. De Sanctis, F. Altieri, E. D'Aversa, M. Ciarniello, " <i>Simulations of Enceladus' plumes with a Smoothed Particle Hydrodynamics model</i> ", JUICE workshop, Les Houches, France.
7-12 Sept. 2025	M. Teodori , L. Maggioni, G. Magni, M. Formisano, M. C. De Sanctis, F. Altieri, E. D'Aversa, M. Ciarniello, " <i>The evolution of a post-impact plume on the surface of the Moon through a Smoothed Particle Hydrodynamics approach</i> ", EPSC-DPS Joint Meeting 2025, Helsinki, Finland, EPSC-DPS2025-323 , DOI: 10.5194/epsc-dps2025-323.
7-12 Sept. 2025	M. Teodori , L. Maggioni, G. Magni, M. Formisano, M. C. De Sanctis, F. Altieri, E. D'Aversa, M. Ciarniello, " <i>Simulations of Enceladus' plumes with a Smoothed Particle Hydrodynamics model</i> ", EPSC-DPS Joint Meeting 2025, Helsinki, Finland, EPSC-DPS2025-308 , DOI: 10.5194/epsc-dps2025-38.
1-5 Sept. 2025	M. Teodori , O. Straniero, M. Merafina, " <i>Primordial segregation in Multiple Populations through N-body simulations and dynamical models</i> ", Bridging scales: star clusters and their hosts from the Local to the high-z Universe, Matera, Italy.
23-27 June 2025	M. Teodori , O. Straniero, M. Merafina, " <i>Dynamical models' view on Globular Clusters and their Multiple Populations</i> ", European Astronomical Society Annual Meeting, Cork, Ireland.
16-20 June 2025	M. Teodori , O. Straniero, M. Merafina, " <i>Dynamical models for exploring the energy equipartition degree and mass segregation in Globular Clusters and their Multiple Populations</i> ", STARS II: Current Challenges, Upcoming Solutions, INAF - OAS, Bologna, Italy.

19-23 Aug. 2024

M. Teodori, O. Straniero and M. Merafina, "[Measuring energy equipartition in Globular Clusters with dynamical models](#)", MODEST-24: Exploring Dense Stellar Systems Across Cosmic Time, Nicolaus Copernicus Astronomical Center, Warsaw, Poland.

16-21 June 2024

M. Teodori, G. Magni, M. Formisano, L. Maggioni, M. C. De Sanctis, F. Altieri, "*Volatiles emission from the Moon's surface: a Smoothed Particle Hydrodynamics approach*", European Lunar Symposium 2024, Dumfries and Galloway, Scotland, United Kingdom. Awarded of a Travel Grant.

8-12 May 2023

M. Teodori, G. Magni, M. Formisano, M. C. De Sanctis, F. Altieri, "*Volatiles emission from a cavity on a planetary surface using smoothed particle hydrodynamics*", Biennial European Astrobiology Conference BEACON 2023, La Palma & Teneguia Princess Hotel on Fuencaliente, La Palma Island, Canary Islands, Spain. Awarded of a financial support for accommodation.

Abstract as co-author

2026

L. Maggioni, **M. Teodori**, G. Magni, M. Formisano, M. C. De Sanctis, F. Altieri, M. Formisano, "*Smoothed Particle Hydrodynamics modeling of volatile emissions: drill-induced releases in the Martian subsurface*", Ma_MISS Team / IDS Subsurface Joint Meeting, Rome, Italy. *Oral presentation*.

2026

G. Rinaldi, J. Noonan, **M. Teodori**, L. Maggioni, M. Formisano "*Analysis of Dust and Gas emission at 67P/Churyumov-Gerasimenko*", EGU General Assembly 2026, Vienna, Austria, [EGU26-20540](#). *Poster presentation*.

2026

M. Formisano, **M. Teodori**, P. Hayne, M. Ciarniello, G. Magni, "*Modeling thermal convection within Europa's icy layer: constraints on shell thickness*", CNSP26, Reggio Emilia, Italy. *Poster presentation*

2026

G. Magni, **M. Teodori**, M. Formisano, L. Maggioni, M. C. De Sanctis, P. O. Hayne, "*An exogenic contribution to lunar water ice: a cometary post-impact plume through Smoothed Particle Hydrodynamics*", CNSP26, Reggio Emilia, Italy. *Oral presentation*.

2025

G. Rinaldi, J. W. Noonan, **M. Teodori**, L. Maggioni, A. S. Jindal, M. Formisano, G. Magni, "*The analysis of dust and gas emission at 67P/Churyumov-Gerasimenko sheds light on cometary activity*", EPSC-DPS 2025, Helsinki, Finland, [EPSC-DPS2025-1556](#), DOI: 10.5194/epsc-dps2025-1556. *Oral presentation*.

2025

M. Formisano, A. Raponi, **M. Teodori**, S. Bertoli, M. Ciarniello, S. De Angelis, M. C. De Sanctis, G. Filacchione, A. Frigeri, L. Maggioni, G. Magni, "*Thermal role of self-heating and surface roughness in micro cold trap stability at the Lunar Poles*", EPSC-DPS 2025, Helsinki, Finland, [EPSC-DPS2025-1411](#), DOI: 10.5194/epsc-dps2025-1411. *Oral presentation*.

2025

L. Maggioni, **M. Teodori**, G. Magni, F. Altieri, M. C. De Sanctis, F. Altieri, M. Formisano, "*Gas mixing at Martian atmospheric conditions through a smoothed particle hydrodynamics approach*", EPSC-DPS 2025, Helsinki, Finland, [EPSC-DPS2025-613](#), DOI: 10.5194/epsc-dps2025-613. *Poster presentation*.

2025

L. Maggioni, M. C. De Sanctis, F. Altieri, M. Formisano, **M. Teodori**, G. Magni, "*Smoothed Particle Hydrodynamics model for volatile emissions from Mars' subsurface triggered by the drill onboard the ESA Rosalind Franklin rover*", EPSC-DPS 2025, Helsinki, Finland, [EPSC-DPS2025-527](#), DOI: 10.5194/epsc-dps2025-527. *Poster presentation*.

2024

L. Maggioni, **M. Teodori**, G. Magni, M. Formisano, M. C. De Sanctis, F. Altieri, "*Volatiles emission from planetary fractures through a Smoothed-Particle-Hydrodynamics approach: the case of Mars*", Europlanet Science Congress 2024, Berlin, Germany, 8–13 Sep 2024, [EPSC2024-312](#), DOI: 10.5194/epsc2024-312.

2024

M. Formisano, M. C. De Sanctis, A. Frigeri, S. Boazman, D. Heater, S. De Angelis, M. Ferrari, G. Magni, **M. Teodori**, F. Altieri, "*Numerical analysis of a south pole lunar region through a 3-D FEM model: the case of the PROSPECT landing site*", European Lunar Symposium 2024, Dumfries and Galloway, Scotland, United Kingdom.

2023	M. Formisano, G. Magni, M. Teodori , M. C. De Sanctis, A. Raponi, M. Ciarniello, G. Filacchione, O. Aharonson, S. Bertoli, J. Castillo-Rogez, N. Davari, A. Deutsch, C. Federico, A. Frigeri, D. Hurley, " <i>Thermophysical characterization of ice-rich lunar polar regions</i> ", European Lunar Symposium 2023, Padova, Italy.
Participation to workshops	
May 2026	Volatile Emissions Workshop 2026, Rome, Italy.
Feb. 2026	MAJIS Science Team Meeting #10, Institute of Space Research German Aerospace Center (DLR), Berlin, Germany.
May 2025	Volatile Emissions Workshop 2025, Rome, Italy.
March 2025	11th Dust Workshop, Venice, Italy. Remote participation.
Nov. 2024	MAJIS Science Team Meeting #9, ASI, Rome, Italy.
March 2024	International Team Meeting at International Space Science Institute (ISSI), Bern, Switzerland.
Nov. 2022	G11 Workshop, Physics Department, University of Rome "La Sapienza".
Funded projects	
2024	Participation as Co-I in the INAF MiniGrant for the project <i>PLUMES: Planetary fractures Lagrangian simUlations for Multi-component EmissionS</i> , PI: Dr. Emiliano D'Aversa. Funding Institute: INAF . Role: development and refinement of the SPH model.
2024	Participation in the INAF Theory Grant for the project <i>Thermophysical modeling of Permanently Shadowed Regions (PSRs) on Moon and Mercury</i> , PI: Dr. Michelangelo Formisano (IAPS). Funding Institute: INAF . Role: refinement of the SPH code and its application.
2023-2024	Participation in the project " <i>Thermophysical characterization of ice-rich areas on the surface of specific planetary bodies: conditions for the formation of a transient exosphere</i> ". PI: Dr. Michelangelo Formisano. Funding Institute: International Space Science Institute (ISSI) . Role: development of SPH codes for volatiles emission from planetary surfaces and subsurfaces.
2022	Participation in the project " <i>Stellar evolution and dynamical evolution in Globular Clusters: theoretic development and N-body simulations</i> ". PI: Prof. Marco Merafina, Funding Institute: University of Rome La Sapienza . Role: development of multi-mass dynamical models.
Computational resources	
2025	1500 GPU hours assigned for the project <i>N-body simulations for segregated Globular Clusters: tests with GPUs</i> , Role: PI . Funding institute: INAF - USC VIII - Computing within the <i>Call #6 "a sportello"</i> .
2025	400k core hours assigned for the project <i>N-body simulations for primordially segregated Multiple Populations in Globular Clusters</i> , Role: PI , simulations setup and analysis of results. Funding institute: INAF - USC VIII - Computing within the <i>Call #5 for Computing and long-term archiving resources</i> .
Organization of workshops	
2026 (upcoming)	Convener for the TP10 Planetary Cryospheres: Ices in the Solar System session of the EPSC2026.
2026	Second edition of the Volatile Emissions Workshop, promoting the mobility of young researchers and PhD students and funded by SISP-AC .
21-22 May 2025	Organization of a workshop to gather Italian expertise on the study of volatiles emission from planetary surfaces, to be addressed with numerical models. The workshop promotes the mobility of young researchers and PhD students and is funded by SISP-AC .

Peer-review activity

Since 2025

Ad Hoc reviewer for the international journal *Icarus*.

Code/software development

Language

Description

C, Cython, Python

SPH4VE – Smoothed Particle Hydrodynamics for Volatile Emissions: SPH code for simulating volatiles emission from planetary surfaces, tracing their dynamical and thermal evolution.

Fortran

KingMM: code to obtain equilibrium solution of the King-like multimass dynamical model, to describe Globular Clusters 3D and 2D observables (structural, spatial, kinematic and thermodynamic properties).

C

Development of a customized version of **MCLUSTER** code, to set up N-body initial conditions for Multiple Population, in the case of single and multi-mass dynamical models.

Python

Codes for analyzing and fitting globular clusters observational data concerning kinematical properties related to energy equipartition (HST proper motion) as well as surface brightness profiles.

Python

Codes for analyzing results of N-body simulations.

Python

Codes for visualization and analysis of SPH simulations results.

Fortran

Code for solving equilibrium configuration of the single-mass King-model, produce results concerning structural properties, spatial distribution, internal kinematics and stability of Globular Clusters (Master Thesis).

C

Code for carbon ignition curve and their analysis (Bachelor Thesis).

Coding/software experience

Programming languages

C intermediate level (6 yrs, Bachelor's degree thesis and courses, PhD project), **Fortran** intermediate level (4 yrs, Master thesis, PhD project), **MATLAB** (for programming) basic level (PhD course) and **Python** intermediate level (4 yrs, INAF experiences, PhD project and Post-doc).

Professional skills

Intermediate experience (4 yrs) with parallel codes and simulations: **PySPH** for hydrodynamical SPH simulations, **MCLUSTER** and **NBODY6++** respectively for setting initial conditions and run N-body simulations of Globular Clusters and their Multiple Populations.

Data analysis and visualization

Experience with **MATLAB** (6 yrs) and **Python** (4 yrs) acquired during university courses, thesis work, INAF experiences, PhD project and Post-doc.

Document drafting

Experience of 10 yrs with **L^AT_EX**, in particular for scientific reports, thesis, abstracts, papers and proposal drafting.

Remote control

Intermediate knowledge (4 yrs) of remote connection to servers for running numerical simulation using SSH and SFTP protocol or by using a remote desktop software (AnyDesk, Splashtop and TeamViewer).

Website

Basic knowledge of HTML (website customization).

Others

Intermediate experience in **Office automation** packages, in particular with software for presentation, document elaboration and spreadsheets, refined from Italian secondary school up to now (around 14 yrs experience). Basic ability in managing videoconferencing.

PhD schools

25-30 January 2026

[Winter school at Les Houches: the exploration of Jupiter and its moons by ESA's JUICE mission](#), Les Houches, France.

24 June - 5 July 2024

INAF - [Course in Computing and High Performance Computing in Astronomy & Astrophysics](#), Bologna, Italy.

2-6 Oct. 2023

INAF - [Scientific Communication in Astronomy School](#), Bertinoro, Italy.

Seminars

12th June 2024

16th Feb. 2023

IAPS Seminar, "*Smoothed Particle Hydrodynamics: simulation of volatiles emission from planetary surfaces*".

INAF-OAAb colloquia, "*The interconnection between multi-mass dynamical models and multiple populations in Globular Clusters*".

Teaching experiences

Supervision

Co-supervision of master's thesis work by F. Di Lernia (INAF-IAPS) and F. Soligo (Department of Physics, Sapienza University).

Tutoring

Tutoring of high school students in Math and Physics.

Personal skills

Mother tongue

Other languages

*Self-assessment
European level**

English

Italian

Understanding		Speaking		Writing
Listening	Reading	Spoken interaction	Spoken production	
B1 Independent user	B1 Independent user	B1 Independent user	B1 Independent user	B1 Independent user

* Common European Framework of Reference (CEF) level